

# Computational Mechanics

## 2. Predicting the future

### Solutions

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#### Instructions

These notes reproduce each exercise before its solution. Equivalent arguments and notation should receive full credit when the mathematical reasoning is correct.

### Exercise 1

#### From statistics to partitions

Let  $u_1, \dots, u_6 \in \mathcal{X}_+^*$  be six possible histories. Suppose that they induce the following conditional distributions over the entire future:

| history             | $u_1$ | $u_2$ | $u_3$ | $u_4$ | $u_5$ | $u_6$ |
|---------------------|-------|-------|-------|-------|-------|-------|
| future distribution | $P$   | $P$   | $P$   | $Q$   | $Q$   | $S$   |

where  $P$ ,  $Q$ , and  $S$  are pairwise distinct.

Three statistics  $R_A$ ,  $R_B$ , and  $R_C$  induce the partitions

$$\Pi_A = \{\{u_1, u_2\}, \{u_3\}, \{u_4\}, \{u_5\}, \{u_6\}\},$$

$$\Pi_B = \{\{u_1, u_2, u_3\}, \{u_4, u_5\}, \{u_6\}\},$$

$$\Pi_C = \{\{u_1, u_2, u_3, u_4\}, \{u_5\}, \{u_6\}\}.$$

Recall that a statistic is sufficient for prediction when histories assigned the same value induce the same distribution over the future.

- What does it mean for two histories to belong to the same cell of the partition induced by a statistic? Illustrate your answer using  $u_1$  and  $u_2$  in  $\Pi_A$ .
- Determine which of  $R_A$ ,  $R_B$ , and  $R_C$  are sufficient. If a statistic is not sufficient, give two histories witnessing the failure.
- Which cells of  $\Pi_A$  can be merged without losing sufficiency? Perform all such merges and identify the resulting partition.
- Explain why no two cells of  $\Pi_B$  can be merged while preserving sufficiency. What does this show about  $R_B$ ?
- Prove that every refinement of a sufficient partition is also sufficient. Is every coarsening of a sufficient partition necessarily sufficient? Use  $\Pi_A$ ,  $\Pi_B$ , and  $\Pi_C$  to illustrate your answer.

(f) **Extension.** Consider the partition

$$\Pi_D = \{\{u_1\}, \{u_2, u_3\}, \{u_4, u_5\}, \{u_6\}\}.$$

Show that  $\Pi_A$  and  $\Pi_D$  are both sufficient but that neither refines the other. How can both nevertheless refine the same minimal sufficient partition?

### Solution.

- (a) Two histories belong to the same cell when they receive the same summary value. In particular,  $R_A(u_1) = R_A(u_2)$ .
- (b) Both  $R_A$  and  $R_B$  are sufficient. Every cell of  $\Pi_A$  and every cell of  $\Pi_B$  contains histories carrying only one of the distributions  $P$ ,  $Q$ , or  $S$ .

The statistic  $R_C$  is not sufficient. Its first cell contains  $u_1$  and  $u_4$ , for example, but  $u_1$  induces  $P$  while  $u_4$  induces  $Q$ . Since  $P \neq Q$ , these histories cannot be assigned the same value by a sufficient statistic.

- (c) The cell  $\{u_1, u_2\}$  can be merged with  $\{u_3\}$  because all three histories induce  $P$ . Similarly,  $\{u_4\}$  can be merged with  $\{u_5\}$  because both histories induce  $Q$ . No cell can be merged with  $\{u_6\}$ , and no  $P$ -cell can be merged with a  $Q$ -cell. After all permissible merges, the resulting partition is

$$\{\{u_1, u_2, u_3\}, \{u_4, u_5\}, \{u_6\}\} = \Pi_B.$$

- (d) The three cells of  $\Pi_B$  correspond respectively to the three distinct future distributions  $P$ ,  $Q$ , and  $S$ . Merging any two cells would therefore place histories with different future distributions in the same cell and destroy sufficiency. Hence  $R_B$  is minimal sufficient.
- (e) Let  $\Pi'$  refine a sufficient partition  $\Pi$ . Any two histories in the same cell of  $\Pi'$  also belong to the same cell of  $\Pi$ . Since  $\Pi$  is sufficient, those histories induce the same future distribution. Therefore  $\Pi'$  is sufficient.

In the example,  $\Pi_A$  refines  $\Pi_B$ , and both are sufficient. A coarsening need not remain sufficient:  $\Pi_C$  is obtained from the sufficient partition  $\Pi_A$  by merging its cells  $\{u_1, u_2\}$ ,  $\{u_3\}$ , and  $\{u_4\}$ . The resulting cell contains histories inducing both  $P$  and  $Q$ , so  $\Pi_C$  is not sufficient.

- (f) Every cell of  $\Pi_D$  lies within one predictive class, so  $\Pi_D$  is sufficient. The same was established for  $\Pi_A$ .

The cell  $\{u_1, u_2\}$  of  $\Pi_A$  is split between  $\{u_1\}$  and  $\{u_2, u_3\}$  in  $\Pi_D$ , so  $\Pi_A$  does not refine  $\Pi_D$ . Conversely,  $\{u_2, u_3\}$  in  $\Pi_D$  is split between  $\{u_1, u_2\}$  and  $\{u_3\}$  in  $\Pi_A$ , so  $\Pi_D$  does not refine  $\Pi_A$ . Both partitions nevertheless refine  $\Pi_B$ , because every one of their cells is contained in one of the three predictive classes represented by  $P$ ,  $Q$ , and  $S$ .

## Exercise 2

### Causal states are minimal sufficient statistics

For  $x, x' \in \mathcal{X}_+^*$ , define the relation  $\sim$  by

$$x \sim x' \iff \Pr(X_{|x|+1:|x|+|y|} = y \mid X_{1:|x|} = x) = \Pr(X_{|x'|+1:|x'|+|y|} = y \mid X_{1:|x'|} = x')$$

for every  $y \in \mathcal{X}^*$ . Define the causal state of  $x$  by

$$\epsilon(x) := [x]_{\sim},$$

and write  $\mathcal{S} := \epsilon(\mathcal{X}_+^*)$  for the set of causal states.

(a) Prove that  $\sim$  is an equivalence relation on  $\mathcal{X}_+^*$  by establishing each of the following:

- (i) reflexivity:  $x \sim x$  for every  $x \in \mathcal{X}_+^*$ ;
- (ii) symmetry: if  $x \sim x'$ , then  $x' \sim x$ ;
- (iii) transitivity: if  $x \sim x'$  and  $x' \sim x''$ , then  $x \sim x''$ .

(b) Prove that  $\epsilon$  is sufficient for prediction.

(c) Let  $R : \mathcal{X}_+^* \rightarrow \mathcal{R}$  be any sufficient statistic. Prove that

$$R(x) = R(x') \implies \epsilon(x) = \epsilon(x').$$

Deduce that every sufficient partition refines the causal-state partition.

(d) Using parts (b) and (c), conclude that  $\epsilon$  is a minimal sufficient statistic.

(e) **Extension.** Let  $T$  be another minimal sufficient statistic. Prove that  $T$  and  $\epsilon$  induce the same partition of  $\mathcal{X}_+^*$ . Deduce that there is a bijection

$$\psi : \mathcal{S} \rightarrow \text{im}(T)$$

satisfying  $T = \psi \circ \epsilon$ .

### Solution.

- (a) (i) For every  $x \in \mathcal{X}_+^*$  and every continuation  $y$ , the conditional probability after  $x$  equals itself. Hence  $x \sim x$ , so  $\sim$  is reflexive.
- (ii) If  $x \sim x'$ , then the two continuation probabilities are equal for every  $y$ . Reversing the equality shows that  $x' \sim x$ , so  $\sim$  is symmetric.
- (iii) Suppose  $x \sim x'$  and  $x' \sim x''$ . For every  $y$ , the continuation probability after  $x$  equals that after  $x'$ , which equals that after  $x''$ . Thus  $x \sim x''$ , so  $\sim$  is transitive.

Therefore  $\sim$  is an equivalence relation.

- (b) If  $\epsilon(x) = \epsilon(x')$ , then  $x$  and  $x'$  belong to the same equivalence class, so  $x \sim x'$ . By the definition of  $\sim$ , the two histories induce the same probability for every finite continuation  $y$ . This is precisely the definition of sufficiency for prediction, so  $\epsilon$  is sufficient.
- (c) Suppose  $R(x) = R(x')$ . Since  $R$  is sufficient,  $x$  and  $x'$  induce the same probability for every finite continuation. Hence  $x \sim x'$ , and equivalent histories have the same equivalence class:  $\epsilon(x) = \epsilon(x')$ .

Thus any two histories in the same cell induced by  $R$  also lie in the same causal state. Every cell induced by  $R$  is therefore contained in a causal state, so the partition induced by  $R$  refines the causal-state partition. Since  $R$  was arbitrary, every sufficient partition has this property.

- (d) Part (b) shows that  $\epsilon$  is sufficient. Part (c) shows that its partition is refined by the partition of every sufficient statistic. Hence the causal-state partition is the coarsest sufficient partition, so  $\epsilon$  is a minimal sufficient statistic.
- (e) Since  $T$  is sufficient, part (c), with  $R = T$ , shows that the partition induced by  $T$  refines the causal-state partition. Since  $T$  is minimal and  $\epsilon$  is sufficient, the causal-state partition must also refine the partition induced by  $T$ . The two partitions are therefore equal.

Define  $\psi(\epsilon(x)) := T(x)$ . Equality of the two partitions shows that this definition is independent of the representative  $x$  and that distinct causal states receive distinct  $T$ -values. It also reaches every value in  $\text{im}(T)$ . Hence  $\psi$  is a bijection.

## Exercise 3

### From finite generators to causal states

In this exercise,  $S_t$  denotes the hidden state immediately before the transition that emits  $X_{t+1}$ . All parameters lie strictly between 0 and 1.

- (a) **The Zero–One–Random Process.** Let  $S_0$  be uniformly distributed over  $\{s_0, s_1, s_R\}$ . The process has transitions

$$s_0 \xrightarrow{0:1} s_1, \quad s_1 \xrightarrow{1:1} s_R, \quad s_R \begin{cases} \xrightarrow{0:1-p} s_0, \\ \xrightarrow{1:p} s_0. \end{cases}$$

An edge label  $x : q$  means that the process emits  $x$  and follows the edge with probability  $q$ .

- (i) List all possible histories of length two. For each history, determine whether it uniquely identifies the current hidden phase. When it does, identify that phase.
- (ii) For each ambiguous length-two history, determine which additional observations resolve the ambiguity. Identify the resulting phase and explain why it remains known after every subsequent observation.
- (iii) Assume that  $\epsilon(\emptyset)$ ,  $\epsilon(0)$ ,  $\epsilon(1)$ , and  $\epsilon(10)$  are pairwise distinct. Show that the causal states have shortest-length representatives

$$\emptyset, \quad 0, \quad 1, \quad 10, \quad 00, \quad 01, \quad 11.$$

Explain why no additional causal states arise.

- (iv) Under the assumption in part (iii), draw the automaton associated with the causal-state dynamics.
- (b) **The Even Process.** Let  $S_0$  be uniformly distributed over  $\{A, B\}$  and consider the generator

$$A \xrightarrow{0:q} A, \quad A \xrightarrow{1:1-q} B, \quad B \xrightarrow{1:1} A.$$

Thus every run of 1s between successive 0s has even length.

- (i) For each of the length-one histories 0 and 1, determine whether the current hidden state can be identified exactly.
- (ii) Starting from each ambiguous history in part (i), consider its possible one-symbol extensions. Which extensions resolve the ambiguity, and which remain ambiguous? You may use the fact that  $\epsilon(\emptyset) = \epsilon(11)$ . What does this imply for histories of the form  $1^n$ ?
- (iii) Assume additionally that  $\epsilon(\emptyset) \neq \epsilon(1)$ . Show that the causal states have shortest representatives

$$\emptyset, \quad 1, \quad 0, \quad 01,$$

and explain why there are no additional causal states. You may distinguish causal states using possible and impossible future continuations.

- (iv) Using  $\epsilon(\emptyset) = \epsilon(11)$  and assuming  $\epsilon(\emptyset) \neq \epsilon(1)$ , draw the automaton associated with the causal-state dynamics.
- (c) **The Simple Nonunifilar Source.** Let  $S_0$  be uniformly distributed over  $\{A, B\}$  and consider

$$A \xrightarrow{0:1/2} A, \quad A \xrightarrow{0:1/2} B, \quad B \xrightarrow{0:1/2} B, \quad B \xrightarrow{1:1/2} A.$$

- (i) Show that observing a 1 identifies the new hidden state as  $A$ .
- (ii) Starting from  $A$ , show that there are  $n+1$  hidden paths that emit  $0^n$ , each with probability  $2^{-n}$ . How many of these paths can subsequently emit 1? Hence calculate

$$\Pr(X_{n+2} = 1 \mid X_{1:n+1} = 10^n).$$

- (iii) Show that this probability is different for every  $n \geq 0$ . Conclude that the histories

$$1, 10, 100, \dots$$

belong to distinct causal states and that the process has infinitely many causal states.

### Solution.

- (a) (i) All four length-two histories are possible. Their resulting hidden phases are

| $x_{1:2}$ | 00    | 01    | 10             | 11    |
|-----------|-------|-------|----------------|-------|
| $S_2$     | $s_1$ | $s_R$ | $\{s_0, s_1\}$ | $s_0$ |

Thus 00, 01, and 11 identify the phase exactly, while 10 is ambiguous.

- (ii) From the phases compatible with 10, only  $s_0$  can emit 0, and only  $s_1$  can emit 1. Consequently, 100 identifies the new phase as  $s_1$ , while 101 identifies it as  $s_R$ . From each phase, the emitted symbol determines the next phase, so every later observation preserves synchronization.
- (iii) The histories 00, 01, and 11 identify  $s_1$ ,  $s_R$ , and  $s_0$ , respectively. These phases induce distinct next-token distributions:  $s_1$  emits only 1,  $s_0$  emits only 0, and  $s_R$  can emit either symbol. Hence they give three distinct causal states.

They are also distinct from the four states assumed in the question. This can be seen without calculating probabilities by comparing possible length-two continuations:

| representative | possible length-two continuations |
|----------------|-----------------------------------|
| $\emptyset$    | $\{00, 01, 10, 11\}$              |
| 0              | $\{01, 10, 11\}$                  |
| 1              | $\{00, 01, 10\}$                  |
| 10             | $\{01, 10, 11\}$                  |
| 00             | $\{10, 11\}$                      |
| 01             | $\{00, 10\}$                      |
| 11             | $\{01\}$                          |

The only repeated continuation set belongs to 0 and 10, whose causal states are assumed to be distinct. Finally, every history of length at least three identifies the phase by part (ii), and so belongs to one of the states represented by 00, 01, and 11. No further causal states arise.

- (iv) Appending each possible symbol gives the causal-state update:

| state                 | 0              | 1              |
|-----------------------|----------------|----------------|
| $\epsilon(\emptyset)$ | $\epsilon(0)$  | $\epsilon(1)$  |
| $\epsilon(0)$         | $\epsilon(00)$ | $\epsilon(01)$ |
| $\epsilon(1)$         | $\epsilon(10)$ | $\epsilon(11)$ |
| $\epsilon(10)$        | $\epsilon(00)$ | $\epsilon(01)$ |
| $\epsilon(00)$        | —              | $\epsilon(01)$ |
| $\epsilon(01)$        | $\epsilon(11)$ | $\epsilon(11)$ |
| $\epsilon(11)$        | $\epsilon(00)$ | —              |

Here a dash denotes an impossible extension. Drawing one node for each row and the indicated symbol-labelled edges gives the required automaton.

- (b) (i) The history 0 identifies the current state as  $A$ : only  $A$  can emit 0, and this transition returns to  $A$ . The history 1 is ambiguous. If  $S_0 = A$ , it leads to  $B$ , while if  $S_0 = B$ , it leads to  $A$ .
- (ii) From the two states compatible with the history 1, only  $A$  can emit 0. Therefore 10 resolves the ambiguity and identifies the new state as  $A$ . The extension 11 remains ambiguous. Using the given equality and recursively updating after pairs of 1s gives

$$\epsilon(1^{2k}) = \epsilon(\emptyset), \quad \epsilon(1^{2k+1}) = \epsilon(1), \quad k \geq 0.$$

- (iii) If a history contains a 0, its final 0 synchronizes the generator to  $A$ . An even number of trailing 1s then leaves it in  $A$ , represented by 0, while an odd number leaves it in  $B$ , represented by 01. If the history contains no 0, part (ii) shows that its causal state is represented by  $\emptyset$  or 1, according to the parity of its length. Thus no additional causal states arise.

These four states are distinct. The state represented by 01 cannot emit 0, whereas the other three can. The continuation 10 is impossible after 0 but possible after  $\emptyset$  and 1. Finally,  $\epsilon(\emptyset) \neq \epsilon(1)$  by assumption. Hence the displayed representatives are also shortest.

- (iv) Appending each possible symbol gives the causal-state update:

| state                 | 0             | 1                     |
|-----------------------|---------------|-----------------------|
| $\epsilon(\emptyset)$ | $\epsilon(0)$ | $\epsilon(1)$         |
| $\epsilon(1)$         | $\epsilon(0)$ | $\epsilon(\emptyset)$ |
| $\epsilon(0)$         | $\epsilon(0)$ | $\epsilon(01)$        |
| $\epsilon(01)$        | —             | $\epsilon(0)$         |

Here a dash denotes an impossible extension. Drawing one node for each row and the indicated symbol-labelled edges gives the required automaton.

- (c) (i) The only transition that emits 1 starts from  $B$ , and that transition leads to  $A$ . Thus the hidden state immediately after observing 1 is  $A$ .
- (ii) One path remains in  $A$  while emitting all  $n$  zeros. Each of the other  $n$  paths moves from  $A$  to  $B$  on one of the  $n$  transitions and then remains in  $B$ . Thus there are  $n + 1$  paths, each with probability  $2^{-n}$ , and

$$\Pr(0^n \mid S = A) = (n + 1)2^{-n}.$$

The  $n$  paths that end in  $B$  can subsequently emit 1. Including this final transition, each has probability  $2^{-(n+1)}$ , so

$$\Pr(X_{n+2} = 1 \mid X_{1:n+1} = 10^n) = \frac{n2^{-(n+1)}}{(n + 1)2^{-n}} = \frac{n}{2(n + 1)}.$$

- (iii) Since

$$\frac{n}{2(n + 1)} = \frac{1}{2} - \frac{1}{2(n + 1)},$$

this probability is strictly increasing with  $n$ . Hence every pair of histories  $10^n$  and  $10^m$ , with  $n \neq m$ , already has a different next-token distribution. They must lie in different causal states. The source consequently has infinitely many causal states despite its two-state generator.

## Exercise 4

### Why next-token prediction is enough

Recall that

$$\mathcal{X}_+^* := \{x \in \mathcal{X}^* : \Pr(X_{1:|x|} = x) > 0\}$$

is the set of allowable histories.

Consider a model with parameters  $\theta$  satisfying

$$h_n = f_\theta(h_{n-1}, x_n), \quad y_n = g_\theta(h_{n-1}, x_n).$$

Fix the initial hidden state  $h_0$  and suppose that the model predicts the next-token distribution exactly: for every  $n \geq 1$ ,

$$y_n = \Pr(X_{n+1} \mid X_{1:n} = x_{1:n}), \quad \forall x_{1:n} \in \mathcal{X}_+^*.$$

For every nonempty allowable history, define

$$R(x_{1:n}) := (h_{n-1}, x_n).$$

For  $a \in \mathcal{X}$ , define the deterministic update

$$\Phi_a(h, x) := (f_\theta(h, x), a).$$

For a word  $z_{1:j}$ , write

$$\Phi_{z_{1:j}} := \Phi_{z_j} \circ \dots \circ \Phi_{z_1}, \quad \Phi_{z_{1:0}} := \text{id}.$$

- (a) Show that, whenever  $x_{1:n}z_{1:j}$  is allowable,

$$R(x_{1:n}z_{1:j}) = \Phi_{z_{1:j}}(R(x_{1:n})).$$

Hence show that, if  $R(x_{1:n}) = R(x'_{1:m})$  and both extended histories are allowable, then

$$R(x_{1:n}z_{1:j}) = R(x'_{1:m}z_{1:j}).$$

- (b) Let  $R(x_{1:n}) = R(x'_{1:m})$  and fix  $z_{1:k} \in \mathcal{X}^k$ . As the symbols of  $z_{1:k}$  are read from left to right, use part (a) and exact next-token prediction to show that for each  $0 \leq j < k$  such that the prefixes through  $z_{1:j}$  are allowable,

$$\begin{aligned} \Pr(X_{n+j+1} = z_{j+1} \mid X_{1:n+j} = x_{1:n}z_{1:j}) \\ = \Pr(X_{m+j+1} = z_{j+1} \mid X_{1:m+j} = x'_{1:m}z_{1:j}). \end{aligned}$$

Deduce that either the same first zero-probability extension is encountered after both histories, or every prefix of  $z_{1:k}$  is allowable after both histories.

- (c) In the case where every prefix is allowable, use the conditional chain rule to write

$$\Pr(X_{n+1:n+k} = z_{1:k} \mid X_{1:n} = x_{1:n}) = \prod_{j=0}^{k-1} \Pr(X_{n+j+1} = z_{j+1} \mid X_{1:n+j} = x_{1:n}z_{1:j}).$$

Write the analogous product after  $x'_{1:m}$  and compare its factors. Deal separately with the zero-probability case from part (b), and conclude directly that

$$R(x_{1:n}) = R(x'_{1:m}) \implies$$

$$\Pr(X_{n+1:n+k} = z_{1:k} \mid X_{1:n} = x_{1:n}) = \Pr(X_{m+1:m+k} = z_{1:k} \mid X_{1:m} = x'_{1:m}).$$

Hence  $R(x_{1:n}) = (h_{n-1}, x_n)$  is a sufficient statistic for predicting the entire future.

**Solution.**

(a) Appending one symbol  $a$  gives

$$R(x_{1:n}a) = (f_\theta(h_{n-1}, x_n), a) = \Phi_a(R(x_{1:n})).$$

Repeatedly applying this deterministic update along  $z_{1:j}$  gives

$$R(x_{1:n}z_{1:j}) = \Phi_{z_j} \circ \dots \circ \Phi_{z_1}(R(x_{1:n})) = \Phi_{z_{1:j}}(R(x_{1:n})).$$

If  $R(x_{1:n}) = R(x'_{1:m})$  and both extended histories are allowable, applying the same composite function to the equal starting representations gives

$$R(x_{1:n}z_{1:j}) = R(x'_{1:m}z_{1:j}).$$

(b) Suppose that the prefixes through  $z_{1:j}$  are allowable after both histories. Part (a) gives

$$R(x_{1:n}z_{1:j}) = R(x'_{1:m}z_{1:j}).$$

The two representations give the same arguments to  $g_\theta$ , so exact next-token prediction gives

$$\begin{aligned} \Pr(X_{n+j+1} = z_{j+1} \mid X_{1:n+j} = x_{1:n}z_{1:j}) \\ = \Pr(X_{m+j+1} = z_{j+1} \mid X_{1:m+j} = x'_{1:m}z_{1:j}). \end{aligned}$$

Thus the next extension has positive probability after one history exactly when it has positive probability after the other. Following the word from left to right, either both encounter their first zero at the same position or all its prefixes are allowable after both.

(c) If every prefix is allowable, the conditional chain rule gives the product in the question and

$$\begin{aligned} \Pr(X_{m+1:m+k} = z_{1:k} \mid X_{1:m} = x'_{1:m}) \\ = \prod_{j=0}^{k-1} \Pr(X_{m+j+1} = z_{j+1} \mid X_{1:m+j} = x'_{1:m}z_{1:j}). \end{aligned}$$

Part (b) shows that the factors in the two products are equal, so the products are equal. If a first zero-probability extension is encountered instead, part (b) shows that it occurs after both histories; consequently both probabilities of the complete continuation are zero.

Therefore equal values of  $R$  imply equal conditional probabilities for every finite continuation. By the definition of sufficiency,

$$R(x_{1:n}) = (h_{n-1}, x_n)$$

is a sufficient statistic for predicting the entire future.